# Dongsik Yoon

*Curriculum Vitae*
*Last updated October 2026*

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https://stillrot.github.io/

**RESEARCH INTEREST**

Real-World Computer Vision (Action Recognition, Anomaly Detection in CCTV)
Synthetic Data Generation via Generative Models
Edge AI (Resource-Efficient Inference, On-Device Deployment)

**EDUCATION**

***M.S.***, Electrical Engineering — Mar. 2021 - Feb. 2023
**Korea University**, Seoul, Korea
Dissertation: Deep Learning Framework Design via Condition Guided Learning
Advisor: Prof. Hanseok Ko (Intelligent Signal Processing Laboratory, ISPL)

***B.S.***, ICT, Multimedia Engineering — Mar. 2015 - Feb. 2021
**Baekseok University**, Cheonan, Korea
Advisor: Prof. No-Yoon Kwak (Image Processing and Computer Graphics Laboratory, IPCG)
Overall GPA 4.36 / 4.5, Major GPA 4.49 / 4.5

**INDUSTRY EXPERIENCE**

***AI Research Engineer*** — May 2024 - Nov 2026
**HDC LABS**, Seoul, Korea
Research on CCTV-based vision systems for indoor safety (fire detection, elevator analytics)

***AI Research Engineer*** — Jan. 2023 - May 2024
**Metaverse Entertainment**, Seoul, Korea
Generative AI for virtual-influencer content (image editing, voice synthesis)

**PUBLICATIONS**

International Conference

***D. Yoon***, *J. Kim, B. Park, S. Kim, S. Hwang, "Does Synthesized Audio Help Action Recognition Beyond Its Predicted Label?",* ***Conference on Robot Learning Workshop (CoRLW)****, 2026*

***D. Yoon***, *"Ask Before It Tells: Benchmark-to-Robot Body-Cue Transfer for a Question-First Bedside Robot",* ***IEEE/RSJ International Conference on Intelligent Robots and Systems Workshop (IROSW)****, 2026*

*H. Park, J. Kim, J. Kim,* ***D. Yoon***, *"CHOP: Chunkwise Context-Preserving Framework for RAG on Multi Documents",* ***arXiv preprint****, 2026*

***D. Yoon****[*], J. Kim[*], D. Lee[*], "Resource-Efficient RGB-Only Action Recognition for Edge Deployment",* ***arXiv preprint****, 2026*

***D. Yoon**, J. Kim, "From Prompts to Deployment: Auto-Curated Domain-Specific Dataset Generation via Diffusion Models", **IEEE/CVF Winter Conference on Applications of Computer Vision Workshop (WACVW)**, 2026*

***D. Yoon**, J. Kim, "Your Super Resolution Model is not Enough for Tackling Real-World Scenarios", **International Conference on Computer Vision Workshop (ICCVW)**, 2025*

*J. Kim*, Y. Lee*, **D. Yoon***, C. Jung, G. Lee, "An Integrated YOLO and VLM System for Fire Detection in Enclosed Environments", **International Conference on Learning Representations Workshop (ICLRW)**, 2025*

***D. Yoon**, J. Kim, S. Song, Y. Lee, G. Lee, "OverallNet: Scale-Arbitrary Lightweight SR Model for handling 360° Panoramic Images", **SIGGRAPH Asia Posters**, 2024*

*J. Kwak, Y. Li, **D. Yoon**, D. Kim, H. Ko, "Injecting 3D Perception of Controllable NeRF-GAN into StyleGAN for Editable Portrait Image Synthesis", **European Conference on Computer Vision (ECCV)**, 2022*

***D. Yoon**, J. Kwak, Y. Li, D. Han, H. Ko, "DIFAI: Diverse Facial Inpainting using StyleGAN Inversion", **IEEE International Conference on Image Processing (ICIP)**, 2022*

*J. Kwak, Y. Li, **D. Yoon**, D. Han, H. Ko, "Generate and Edit Your Own Character in a Canonical View", **IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)**, AI for Content Creation Workshop, 2022*

***D. Yoon**, J. Kwak, Y. Li, Y. Jin, D. Han, H. Ko, "Reference Guided Image Inpainting using Facial Attributes", **British Machine Vision Conference (BMVC)**, 2021*

*J. Kwak, Y. Jin, Y. Li, **D. Yoon**, D. Kim, H. Ko, "Adverse Weather Image Translation with Asymmetric and Uncertainty-aware GAN", **British Machine Vision Conference (BMVC)**, 2021*

*Y. Li, Y. Jin, J. Kwak, **D. Yoon**, D. Han, H. Ko, "Adaptive Content Feature Enhancement GAN for Multimodal Selfie to Anime Translation", **British Machine Vision Conference (BMVC)**, 2021*

International Journal

*Y. Li, **D. Yoon**, B. Ku, H. Ko, "ConSeisGen: Controllable Synthetic Seismic Waveform Generation", **IEEE Geoscience and Remote Sensing Letters (GRSL)**, 2023 (IF 4.8, SCI-E)*

***D. Yoon**, J. Kim, V. Lorant, S. Kang, "Manipulation of Age Variation Using StyleGAN Inversion and Fine-Tuning", **IEEE Access**, 2023 (IF 3.9, SCI-E)*

***D. Yoon**, Y. Li, B. Ku, H. Ko, "Estimation of Magnitude and Epicentral Distance from Seismic Waves using Deeper CRNN", **IEEE Geoscience and Remote Sensing Letters (GRSL)**, 2023 (IF 5.3, SCI-E)*

## PROJECTS

**In-elevator situational awareness** Jan. 2025 - Feb. 2026

Built a PoC for in-cab anomaly detection (pet detection, occupancy estimation) that triggered a HDC LABS / HDC Hyundai Industrial Development / Hyundai Elevator three-party MOU. Constructed a domain-specific synthetic + real labelling pipeline and a deployment-automation environment.

*Affiliation: HDC LABS*

**AI fire / smoke detection for underground parking lots** May 2024 - Oct. 2025

Designed a synthetic-data pipeline for CCTV-specific fire scenes; trained YOLOv8s + Florence2 VLM detector and migrated to a YOLOv8n + CLIP edge variant for Hailo dual-chip deployment. Demonstrated at Gwangmyeong Central IPARK.

*Affiliation: HDC LABS*

**Designer-facing image generation for virtual influencers** Jan. 2023 - May 2024

Curated Stable Diffusion LoRA workflows and a Gradio-based internal UI used for MAVE: / rina.8k content. Researched a two-stage StyleGAN-based face re-aging framework, published in IEEE Access.

*Affiliation: Metaverse Entertainment*

**Expressive multi-speaker TTS** Jan. 2022 - Dec. 2022

Built reference-audio-conditioned emotion control model for reading-style / dialog-style TTS (MOS +5%, Similarity-MOS +11% over baseline, n=30).

*Funding: Deep Machine Laboratory (DMLAB)*

**Generative virtual earthquake scenarios for the Korean Peninsula** Jan. 2021 - Dec. 2021

Built a GAN-based generator and verification pipeline producing M ≥ 3.0 seismic waveforms; delivered a scenario-authoring software to KMA.

*Funding: Korea Meteorological Administration (KMA)*

## PATENTS

**13 KR patents granted, 4 patent applications in progress (3 KR + 1 US)** 2024 - 2026

Topics include fire detection in enclosed environments, in-elevator pet / occupancy analytics, construction-site hazard detection, domain-specific synthetic data generation, personalized object tracking, and AI-driven home agents.

*See stillrot.github.io/#patents for the full list.*

## ACADEMIC SERVICE

***Reviewer (peer-review)***

ICIP 2026; ICBINB @ ICLR 2026; TAG @ CoRL 2026

***Instructor, K-Digital Training*** | HDC LABS 2025 - 2026

Generative AI for Analytics and Service Development

## GRANTS & AWARDS

**Best Paper Presentation Award** 2020

Korea Multimedia Society (KMS) Summer Annual Conference

**Best Paper Award** 2019

Joint Conference of Korea Digital Contents Society (KDCS) and Korea Institute of Information Technology (KIIT)

**Gold Prize, Undergraduate Paper Competition** 2019

KIIT / KDCS Joint Summer Conference

**Encouragement Award** 2019

Korean Institute of Communications and Information Sciences (KICS) Winter Annual Conference